GG-W-00259b (DSA-DM)
April 26, 1972
INTERIM REVISION OF
GG-W-259a
July 22, 1959

INTERIM FEDERAL SPECIFICATION

WHEEL, BUFFING, CLOTH, DENTAL

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification GG-W-259a.

1. SCOPE

1.1 This specification covers a cloth wheel suitable for mounting on tapered lathe chucks and for buffing dental appliances.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-resistant, (For Sealing Fiberboard Boxes.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

FSC 6520

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:49Z
Check the source to verify that this is the currentversion before use.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-191	Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Auburn, Washington and Washington, D. C.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Muslin. The cloth shall be a high quality, fine, unbleached muslin.

3.1.2 Thread. The thread used for stitching shall be unbleached, soft finish, cotton thread.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:49Z
Check the source to verify that this is the current version before use.

3.2 Construction. Wheel shall be made of plies of muslin stitched together with the centers hardened. The wheel center shall be punched with a hole not less than 1/8 inch nor more than 3/16 inch in diameter for mounting on tapered lathe chucks. The wheel shall be flat across its face. The wheels shall be prefaced by machine to make it ready for immediate mounting on dental lathe chuck. The nap on the surface of the wheel shall be not less than 1/16 inch in depth. Wheels shall conform to the requirements shown in table I.

Table I. Construction

Diameter of wheel (tolerance ± 1/16 inch)	Plies (tolerance + 3, - 0)	Rows of stitching	Diameter of innermost row of stitches (tolerance ± 1/16 inch)	Distance between outside row of stitches to peripheral edge of wheel ± 1/8 inch)
Inches			Inch	Inch
4	30	4	9/16	3/4

3.2.1 Plies. The plies of muslin shall be of uniform thickness and texture. They shall extend to the outer edge of the wheel and shall be turned one-twelfth to one-eighth of a circle so that the warp yarns of one ply are turned 30 to 45 degrees to the warp yarns of the group above and below it. The number of plies in each wheel shall be as shown in table I.

3.2.2 Stitching. Stitching shall be continuous concentric machine sewing and shall conform to the stitching requirements of table I. There shall be a minimum of 4 1/2 stitches and a maximum of 7 stitches to an inch.

3.2.3 Center hardening. The centers of wheels shall be hardened with a suitable hardening agent that will not deteriorate and will prevent the wheel from riding up on the lathe spindle.

3.3 Weight per square yard. The muslin shall weigh 4.1 ± 0.1 ounces per square yard when tested as specified in 4.4.1.

3.4 Thread count. The muslin shall have 80 ± 2 yarns per inch in the warp direction and 92 ± 2 yarns per inch in the filling direction when tested as specified in 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:49Z
Check the source to verify that this is the current version before use.

3.5 Breaking strength. The breaking strength of the thread used for stitching shall be 6.0 pounds minimum when tested as specified in 4.4.3.

3.6 Workmanship. Workmanship shall be first class throughout. The muslin used shall be free from holes, mends, sewed pieces, or other defects. The finished buffing wheels shall be free from any other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the components/materials may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirement.

4.2 Quality conformance inspection. The examinations and tests required to assurance conformance of those items or lots of items to be offered are classified as specified in tables II and IV (see 4.2.1 and 4.3) and table III (see 4.2.2.).

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the current version before use.

4.2.1 Sampling for examination . Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. Unit of product for examination purposes shall be one wheel.

Table II Sampling for Examination

Type Defect	Inspection Level	AOL (Percent Defective)
Major	II	1.0
Minor	II	2.5

4.2.2 Sampling for tests. Sufficient samples of the components (muslin and thread) shall be selected from each lot of raw materials to perform tests as required herein. The acceptance number shall be zero for all sample sizes.

Table III Test Criteria

Characteristic	Requirement	Test Procedure
Weight per square yard (muslin)	3.3	4.4.1
Thread count (muslin)	3.4	4.4.2
Breaking strength (thread)	3.5	4.4.3

4.2.3 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

4.3 Examination. The wheels shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Table IV Classification of Defects

Categories	Defects
Critical:	None defined
Major:	
101	Thread not unbleached, soft finish, cotton thread.
102	Muslin not fine, unbleached muslin.
103	Size of hole in wheel center not as specified.
104	Wheel not flat across its face.
105	Plies of muslin not of uniform thickness and texture.
106	Plies of muslin not turned 1/12 to 1/8 of a circle so that warp yarns of one ply are turned 30 to 45 degrees to the warp yarns of the ply above and below it.
107	Stitches not 4 1/2 to 7 per inch.
108	Centers not hardened.
109	Stitching not continuous, concentric stitching.
110	Construction of wheels not as specified in Table I.
Minor:	
201	Muslin not free from holes, mends, sewed pieces, or other defects.
202	Nap on surface of wheel not less than 1/16 inch in depth.
203	Packaging and packing not as specified.
204	Marking on packaging or pack incorrect, illegible, incomplete, or missing altogether.

4.4 Test procedures. Tests of the component materials used in fabrication of the wheels shall be made as specified herein.

4.4.1 Weight per square yard. Test for weight per square yard of the muslin shall be conducted in accordance with method 5041 of Federal Standard FED-STD-191.

4.4.2 Thread count. Test for thread count of the muslin shall be conducted in accordance with method 5050 of Federal Standard FED-STD-191.

4.4.3 Breaking strength. Test for breaking strength of the thread shall be conducted in accordance with method 4100 of Federal Standard FED-STD-191.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One wheel, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	24 units	360 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:49Z
Check the source to verify that this is the current version before use.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table of PPP-B-636 shall apply for PPP-B-636 box.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table of PPP-B-636 shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commerical containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

6.2 Federal Stock Number (FSN)coverage. This specification covers the following item appearing in the Federal Supply Catalog:

6520-568-4150 WHEEL, BUFFING, CLOTH, 4 inch.

6.3 Limitation. This specification does not cover all types, classes, numbers, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those tungsten carbide excavating burs which are normally procured to meet Federal Government requirements.

6.4 Note. The margins of this specification have not been marked with asterisks to indicate changes from the previous issue due to the extensiveness of changes to this document.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1316

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa. 19101

Defense Supply Agency

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DOD-304

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

GG-W-00259b (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.